Master of Applied IT Group project case: *Sustainable semi supervised learning with synthetic data generation*

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Context

In Research Group Sustainable Data&AI Applications, we focus on solving the challenges when applying advanced technology of artificial intelligence. Often when collecting image data for training a computer vision neural network, the data needs to be labelled, which is time consuming and tedious. Sometimes it is also hard or not possible to collect the training data, while synthetic data can be a supplement. How to provide a software solution to support semi supervised learning to overcome those limitations? And how to validate the results?

Significance

To provide a cost efficient, sustainable data and AI solution to the end user requires deeper research and optimization on the whole pipeline of the machine learning operation.

Existing work and available information

Students who have already obtained software and AI knowledges, and/or have experience on labelling data or generating synthetic data, as well as training an AI model can take a step further towards building a sustainable data&AI solution. They can start with data collected from existing research project such as FlowerPower project, or they can also initiate their own project with self-collected data or online data .

The project goal

The goal of this project is to perform research to achieve cost efficient and sustainable data & AI software solutions with semi supervised learning and with synthetic data.

Research questions

How can semi supervised learning and synthetic data help to achieve cost efficient and sustainable AI software solutions? Possible sub questions can be:

1. *How to create a cost efficient solution for generating and labelling synthetic data?*
2. *How to create a semi supervised learning environment and deployment workflow with the synthetic generated data?*
3. *How to validate the quality of semi supervised learning model and the quality of the synthetic generated data. How to access the overall cost and performance?*

Solution characteristics

This project is explorative research project. A prototype such as jupyter notebook functions as a proof of concept for the proposed solution. Possible requirements are:

- *Easy user interaction might be considered for labelling the data*
- *Full deployment pipeline needs to be created on targeted infrastructure on in the cloud*
- *Solution needs to be future proof and be easily transferred to further development*
- *Always required: The solution needs to be GDPR compliant (by law)*
- *Always required: To enable a smooth project handover, the solution needs to be well documented, and the project code should be of high quality (e.g., readable, maintainable, extensible, etc.)*

Schouten, G.; Michielsen, B.S.H.T.; Gravendeel, B. (2024). The Eindhoven Wildflower Dataset: A Data-centric AI Approach for Monitoring Flowering Plants. *Submitted to PlosOne*
[Eindhoven Wildflower Dataset - Fontys Lectoraat AI en Big Data](https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/U4VQJ6)
(https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/U4VQJ6)